

## Solutions to Problem Solving Class 5: TCP and UDP

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Computer Communications – DE  
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1. It means that the sequence and acknowledgement numbers refer to bytes instead of segments. For example, the value of the ack-field in a segment defines the number of the next byte a party expects to receive.
2. Presumptions: window size = 10000 bytes, previous acknowledgement 22001, receives acknowledgement 24001. The movement of the window can be seen in Figure 1.
3. Data are 16 bytes, length of UDP header is 8 bytes, so the ratio is  $\frac{16}{16+8} = \frac{2}{3}$ .
4. Data are 16 bytes, length of IP header (no options) + UDP header is 28 bytes, so the ratio is  $\frac{16}{28+16} = 0.364$ .
5. Data are 16 bytes, length of TCP header (no options) is 20 bytes, so the ratio is  $\frac{16}{20+16} = \frac{4}{9}$ .
6. Data are 16 bytes, length of IP header (no options) + TCP header (no options) is 40 bytes, so the ratio is  $\frac{16}{16+40} = 0.286$ .
7. UDP is a very simple protocol with minimal overhead. If a process wants to send a small message and does not care much about reliability, it can use UDP. Sending a small message using UDP takes much less interaction between sender and receiver than using TCP. UDP is used in multimedia and multicast applications, such as multiplayer games. If reliability is wanted on the other hand, TCP should be chosen. FTP and Telnet use TCP as transport layer protocol.
8. We Interpret the TCP payload of the IP packets carried by the Ethernet frames (let HRC denote the fields of Header length, Control and Reserved).  
Frame

1: Seq. number: F2D27A29, ack. number: 00000000, window size:

$$4000, \text{HRC: } 7002_{16} = \underbrace{0111}_{Hlen} \overbrace{00000000}^{reserved} \underbrace{0000}_{syn} \underbrace{1}_{syn} 0_2.$$

2: Seq. number: A965AB46, ack. number: F2D27A2A, window size:

$$0BB8, \text{HRC: } 6012_{16} = \underbrace{0110}_{Hlen} \overbrace{00000000}^{reserved} \underbrace{0}_{ack} \underbrace{1}_{ack} 00 \overbrace{1}^{syn} 0_2.$$

3: Seq. number: F2D27A2A, ack. number: A965AB47, window size:

$$4440, \text{HRC: } 5010_{16} = \underbrace{0101}_{Hlen} \overbrace{000000}^{reserved} 0 \underbrace{1}_{ack} 0000_2.$$

4: Seq. number: A965AB47, ack. number: F2D27A2A, window size:

$$0BB8, \text{HRC: } 5018_{16} = \underbrace{0101}_{Hlen} \overbrace{000000}^{reserved} 0 \underbrace{1}_{ack} \underbrace{1}_{psh} 000_2.$$

5: Seq. number: F2D27A2A, ack. number: A965AB4A, window size:

$$443D, \text{HRC: } 5018_{16} = \underbrace{0101}_{Hlen} \overbrace{000000}^{reserved} 0 \underbrace{1}_{ack} \underbrace{1}_{psh} 000_2.$$

6: Seq. number: A965AB4A, ack. number: F2D27A30, window size:

$$0BB8, \text{HRC: } 5018_{16} = \underbrace{0101}_{Hlen} \overbrace{000000}^{reserved} 0 \underbrace{1}_{ack} \underbrace{1}_{psh} 000_2.$$

See Figure 2, for the traffic between sender and receiver.

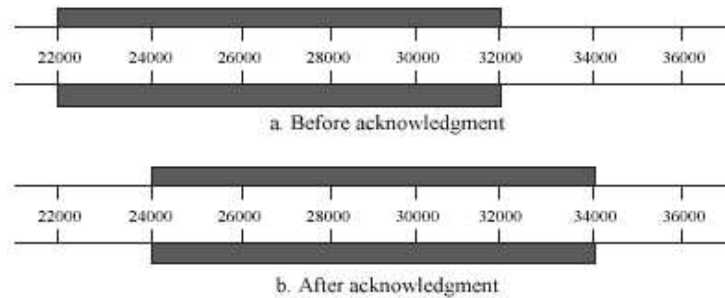


Figure 1: Sliding window of Exercise 2.

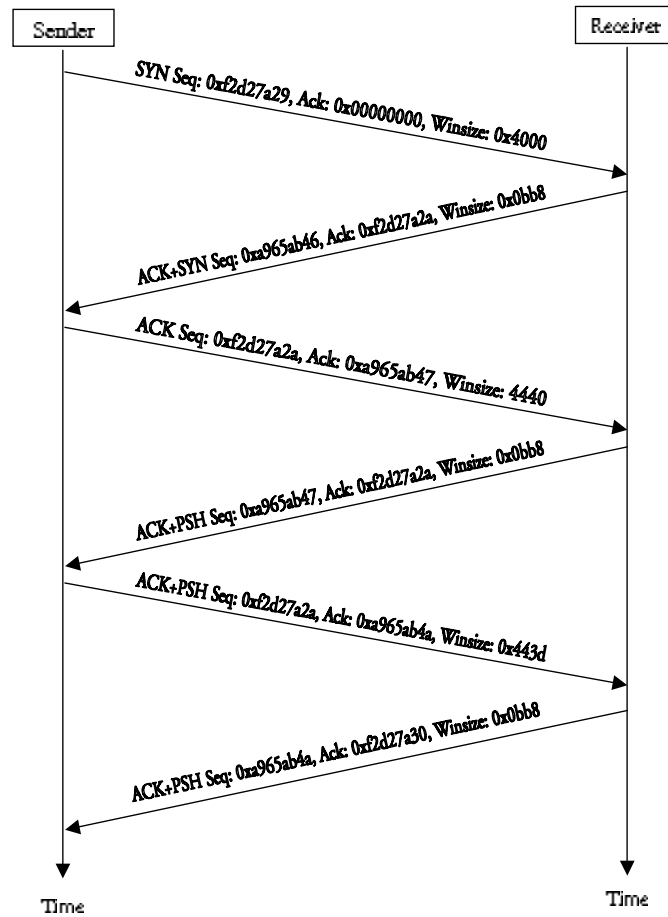


Figure 2: TCP messages for Exercise 8.